Practical No. 9 :-

QUE :-

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

➔

```
~
  GNU nano 8.7
#include <stdio.h>
#include <stdlib.h>

int main() {
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int visited[8] = {0};
    int total_movement = 0;

    printf("Order of servicing:\n");

    for(int i = 0; i < n; i++) {
        int min_dist = 9999;
        int index = -1;

        for(int j = 0; j < n; j++) {
            if(!visited[j]) {
                int dist = abs(head - requests[j]);
                if(dist < min_dist) {
                    min_dist = dist;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        total_movement += min_dist;
        head = requests[index];

        printf("%d -> ", head);
    }

    printf("\nTotal Head Movement = %d\n", total_movement);

    return 0;
}

^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J
```

```
Barkha@DESKTOP-35GK287 MSYS ~
$ nano sstf.c

Barkha@DESKTOP-35GK287 MSYS ~
$ gcc sstf.c -o sstf

Barkha@DESKTOP-35GK287 MSYS ~
$ ./sstf
Order of servicing:
41 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 14 ->
Total Head Movement = 323

Barkha@DESKTOP-35GK287 MSYS ~
$ |
```